ECE251s: Signals and Systems Fundamentals

Spring 2026 Project

Student Activity (10 Marks)

Description:

The aim of this project is to develop a code that does the following:

- Read an audio file from the hard disk of the computer
- Plot the audio signal in time domain
- Calculate the signal energy from time domain
- Find the frequency domain representation of the signal and plot it
- Calculate the signal energy from frequency domain, compare with calculated value in time domain
- Find the frequency that has the maximum magnitude of the signal, f_{peak}.
- Apply an ideal notch filter in the frequency domain with a center around f_{peak} and a bandwidth of your choice around the peak by multiplying the response of the filter by the frequency spectrum of the signal (try 3 different values of filter bandwidth).
- Find the corresponding signal in time domain for the filtered signal and plot it
- Save the filtered signal in time domain as an audio file on the hard disk (of the same format as the original one)

Deliverables:

- A report showing the developed code and plots.
- Students should be ready for a discussion of the project if requested, along with a live test of the code on the students' computer.

Notes:

- The input audio file should be one where each student of the group says his/her name (one audio file for all the group students)
- Students are free to choose the programming language (Python, MATLAB/Octave, C++ … etc.).
- The project should be implemented in groups (not individuals) ideally of **5 students** each
- Useful algorithms: Fast Fourier Transform and its inverse (fft/ifft) / (fftshift)